Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area and perimeter**

A rectangle has the dimensions shown below. Find its area and perimeter.

Area $= 12 \times 7 = 84$ square units. Perimeter $= 2(12 + 7) = 38$ units.

2 **[Foundational]** • **Target: Area and perimeter**

Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

Area $= \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66$.

3 **[Foundational]** • **Target: Area and perimeter**

A triangle has base 15 and height 10. Find its area.

Area $= \frac{1}{2}(15)(10) = 75$.

4 **[Foundational]** • **Target: The Pythagorean theorem**

A right triangle has the two legs shown below. Find the length of the hypotenuse c.

$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

5 **[Foundational]** • **Target: The Pythagorean theorem**

A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

$h = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ feet.

6 **[Foundational]** • **Target: Volume and surface area**

Find the volume of a cylinder with radius 4 and height 9, in terms of π.

$V = \pi r^2 h = \pi(16)(9) = 144\pi$.

7 **[Foundational]** • **Target: Volume and surface area**

Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

$SA = 2(lw + lh + wh) = 2(20 + 15 + 12) = 2(47) = 94$ square units.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle ABC is similar to triangle DEF, with $AB = 8$, $DE = 12$, and $BC = 10$. Find EF.

Scale factor $= \frac{DE}{AB} = \frac{12}{8} = 1.5$. $EF = 10 \times 1.5 = 15$.

9 [Intermediate] • Target: Similar triangles and proportional reasoning

A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.

$\frac{\text{height}}{18} = \frac{5}{3}$, so height $= \frac{18 \times 5}{3} = 30$ feet.

10 [Intermediate] • Target: Angle relationships

Two parallel lines are cut by a transversal. If one angle measures 72°, find its corresponding angle and its co-interior (same-side interior) angle.

The corresponding angle is equal: 72°. The co-interior angle is supplementary: $180^\circ - 72^\circ = 108^\circ$.

11 [Intermediate] • Target: Angle relationships

The three angles of a triangle are in the ratio $2 : 3 : 4$. Find each angle.

The parts sum to $2 + 3 + 4 = 9$, and the total is 180°, so each part is 20°. The angles are 40°, 60°, and 80°.

12 [Intermediate] • Target: Angle relationships

Two angles are complementary, and one is 34° more than the other. Find both angles.

$x + (x + 34) = 90$, so $2x = 56$, giving $x = 28$. The angles are 28° and 62°.

13 [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the distance between the points $(3, -2)$ and $(9, 6)$.

$d = \sqrt{(9-3)^2 + (6-(-2))^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

14 [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.

Midpoint $= \left(\frac{-5+7}{2}, \frac{3+(-9)}{2}\right) = (1, -3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A circle has the radius shown below. Find its area and circumference, in terms of π.

Area $= \pi(8)^2 = 64\pi$. Circumference $= 2\pi(8) = 16\pi$.

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16 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 8.01 Exponentials & Logarithms

Write the equation of a circle with center $(3, -4)$ and radius 6.

$(x-3)^2 + (y+4)^2 = 36$.

17 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 1.01 Heart of Algebra

A circle has the radius and shaded sector shown below. Find the area of the shaded 60° sector, in terms of π.

Sector area $= \frac{60}{360} \times \pi(9)^2 = \frac{1}{6} \times 81\pi = 13.5\pi$.

18 [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 2.01 Quadratics & Polynomials

A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π.

Square area $= 12^2 = 144$. Circle area $= \pi(5)^2 = 25\pi$. Remaining area $= 144 - 25\pi$.

19 [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 3.01 Functions & Graphs

A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.

Garden area $= 300$. Fountain area $\approx \pi(9) \approx 28.26$. Remaining area $\approx 300 - 28.26 \approx 272$ square meters.

20 [Challenging] • Target: A well-designed word problem: designing a park • Cross-Lesson: 4.01 Ratios, Rates & Data

This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of π. (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of π.

(a) The semicircle's diameter equals the 20-meter width, so the radius is 10 meters. (b) Rectangle area $= 30 \times 20 = 600$. Semicircle area $= \frac{1}{2}\pi(10)^2 = 50\pi$. Total area $= 600 + 50\pi$ square meters. (c) The fenced sides are the two 30-meter lengths, the far 20-meter end, and the semicircular arc. The arc length is half the circumference: $\frac{1}{2}(2\pi \times 10) = 10\pi$. Total fence $= 30 + 30 + 20 + 10\pi = 80 + 10\pi$ meters.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Area and perimeter

A rectangle has length 23 and width 5. Find its area and perimeter.

a. $115, 56$ ✓

b. $56, 115$

c. $119, 56$

d. $115, 54$

Area $= 23 \times 5 = 115$ square units. Perimeter $= 2(23 + 5) = 56$ units.

22 [Foundational] • Target: The Pythagorean theorem

A rectangle has length 17 and width 15. Find its area and perimeter.

a. $255, 64$ ✓

b. $64, 255$

c. $259, 64$

d. $255, 62$

Area $= 17 \times 15 = 255$ square units. Perimeter $= 2(17 + 15) = 64$ units.

23 [Foundational] • Target: Volume and surface area

A rectangle has length 27 and width 6. Find its area and perimeter.

a. $162, 66$ ✓

b. $66, 162$

c. $166, 66$

d. $162, 64$

Area $= 27 \times 6 = 162$ square units. Perimeter $= 2(27 + 6) = 66$ units.

24 [Foundational] • Target: Similar triangles and proportional reasoning

A rectangle has length 4 and width 13. Find its area and perimeter.

a. $52, 34$ ✓

b. $34, 52$

c. $52, 32$

d. $56, 34$

Area $= 4 \times 13 = 52$ square units. Perimeter $= 2(4 + 13) = 34$ units.

25 [Foundational] • Target: Angle relationships

A rectangle has length 26 and width 16. Find its area and perimeter.

a. $416, 84$ ✓

b. $416, 82$

c. $420, 84$

d. $84, 416$

Area $= 26 \times 16 = 416$ square units. Perimeter $= 2(26 + 16) = 84$ units.

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26 [Foundational] • Target: Coordinate geometry: distance and midpoint

A rectangle has length 20 and width 15. Find its area and perimeter.

a. $304, 70$

b. $70, 300$

c. $300, 68$

d. $300, 70$ ✓

Area $= 20 \times 15 = 300$ square units. Perimeter $= 2(20 + 15) = 70$ units.

27 [Foundational] • Target: Circles: area, circumference, and equation

A rectangle has length 13 and width 23. Find its area and perimeter.

a. $72, 299$

b. $303, 72$

c. $299, 70$

d. $299, 72$ ✓

Area $= 13 \times 23 = 299$ square units. Perimeter $= 2(13 + 23) = 72$ units.

28 [Foundational] • Target: Composite and shaded regions

A right triangle has legs 3 and 4. Find the length of the hypotenuse c.

a. 5 ✓

b. 7

c. 3

d. 9

$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: designing a park

A right triangle has legs 7 and 24. Find the length of the hypotenuse c.

a. 27

b. 25 ✓

c. 31

d. 23

$c = \sqrt{7^2 + 24^2} = \sqrt{49 + 576} = \sqrt{625} = 25.$

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30 [Intermediate] • Target: Angle Relationships & Polygons

A right triangle has legs 20 and 21. Find the length of the hypotenuse c.

a. 27

b. 31

c. 41

d. 29 ✓

$c = \sqrt{20^2 + 21^2} = \sqrt{400 + 441} = \sqrt{841} = 29.$

31 [Intermediate] • Target: Triangles, Similarity & Special Right Triangles

A right triangle has legs 11 and 60. Find the length of the hypotenuse c.

a. 71

b. 63

c. 61 ✓

d. 59

$c = \sqrt{11^2 + 60^2} = \sqrt{121 + 3600} = \sqrt{3721} = 61.$

32 [Intermediate] • Target: Circles, Arcs & Tangents

A right triangle has legs 18 and 80. Find the length of the hypotenuse c.

a. 98

b. 84

c. 82 ✓

d. 80

$c = \sqrt{18^2 + 80^2} = \sqrt{324 + 6400} = \sqrt{6724} = 82.$

33 [Intermediate] • Target: 3D Surface Area & Volume

A right triangle has legs 27 and 120. Find the length of the hypotenuse c.

a. 125

b. 123 ✓

c. 121

d. 147

$c = \sqrt{27^2 + 120^2} = \sqrt{729 + 14400} = \sqrt{15129} = 123.$

34 [Intermediate] • Target: Area and perimeter

A right triangle has legs 9 and 12. Find the length of the hypotenuse c.

a. 13

b. 21

c. 15 ✓

d. 17

$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$

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35 [Intermediate] • Target: The Pythagorean theorem

Find the volume of a cylinder with radius 6 and height 4, in terms of π.

a. 576π

b. 24π

c. 150π

d. 144π ✓

$V = \pi r^2 h = \pi(36)(4) = 144\pi$.

36 [Intermediate] • Target: Volume and surface area

Find the volume of a cylinder with radius 9 and height 7, in terms of π.

a. 2268π

b. 63π

c. 573π

d. 567π ✓

$V = \pi r^2 h = \pi(81)(7) = 567\pi$.

37 [Intermediate] • Target: Similar triangles and proportional reasoning

Find the distance between the points $(-10, 2)$ and $(-15, -10)$.

a. 13 ✓

b. 11

c. 15

d. 17

$d = \sqrt{((-15)-(-10))^2 + ((-10)-2)^2} = \sqrt{25+144} = \sqrt{169} = 13$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Angle relationships • Cross-Lesson: 2.01 Quadratics & Polynomials

Find the distance between the points $(-5, -9)$ and $(-14, 31)$.

a. 41 ✓

b. 43

c. 39

d. 49

$d = \sqrt{((-14)-(-5))^2 + (31-(-9))^2} = \sqrt{81+1600} = \sqrt{1681} = 41$.

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39 [Challenging] • Target: Coordinate geometry: distance and midpoint • Cross-Lesson: 3.01 Functions & Graphs

A square with side length 19 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π.

a. $365 - 25\pi$

b. $361 - 25\pi$ ✓

c. $361 - 27\pi$

d. 361π

Square area $= 19^2 = 361$. Circle area $= \pi(5)^2 = 25\pi$. Remaining area $= 361 - 25\pi$.

40 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 4.01 Ratios, Rates & Data

A square with side length 16 has a circle of radius 4 removed from its center. Find the remaining area, in terms of π.

a. $260 - 16\pi$

b. 256π

c. $256 - 18\pi$

d. $256 - 16\pi$ ✓

Square area $= 16^2 = 256$. Circle area $= \pi(4)^2 = 16\pi$. Remaining area $= 256 - 16\pi$.

41 [Challenging] • Target: Angle Relationships & Polygons • Cross-Lesson: 5.01 Statistics & Probability

Two angles are complementary. If the measure of one angle is 14° more than three times the other, what is the measure of the smaller angle?

a. 19° ✓

b. 24°

c. 38°

d. 71°

Let the smaller angle be x. Complementary condition:
$x + (3x + 14) = 90 \Rightarrow 4x + 14 = 90 \Rightarrow 4x = 76 \Rightarrow x = 19^\circ$.

42 [Challenging] • Target: Angle Relationships & Polygons • Cross-Lesson: 7.01 Trigonometry & Additional Topics

In ΔABC, the exterior angle at vertex C measures 125°. If one remote interior angle measures 55°, what is the measure of the other remote interior angle?

a. 55°

b. 65°

c. 70° ✓

d. 75°

Exterior angle theorem: exterior angle equals the sum of two remote interior angles:
$55^\circ + x = 125^\circ \Rightarrow x = 70^\circ$.

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43 [Challenging] • Target: Triangles, Similarity & Special Right Triangles • Cross-Lesson: 8.01 Exponentials & Logarithms

Two similar triangles have perimeters of 24 cm and 36 cm. If the area of the smaller triangle is 32 cm^2, what is the area of the larger triangle?

a. 48 cm^2

b. 64 cm^2

c. 72 **cm**^{2} ✓

d. 96 cm^2

Scale factor of sides $= \frac{36}{24} = \frac{3}{2}$. Ratio of areas $= (\frac{3}{2})^2 = \frac{9}{4}$. Larger area $= 32 \times \frac{9}{4} = 72$ cm^2.

44 [Challenging] • Target: Circles, Arcs & Tangents • Cross-Lesson: 1.01 Heart of Algebra

What is the area of a circle whose circumference is 16π?

a. 16π

b. 32π

c. 64π ✓

d. 128π

$2\pi r = 16\pi \Rightarrow r = 8$. Area $= \pi r^2 = \pi(8^2) = 64\pi$.

45 [Challenging] • Target: 3D Surface Area & Volume • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the total volume of a right circular cylinder with radius 5 cm and height 12 cm?

a. 60π cm^3

b. 120π cm^3

c. 300π **cm**^{3} ✓

d. 600π cm^3

Volume $= \pi r^2 h = \pi(5^2)(12) = \pi(25)(12) = 300\pi$ cm^3.

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